

EVENES HARSTAD NARVIK, Norway

WMO: 011830

Lat: 68.491N

Lon: 16.678E

Elev: 84

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	3.5	-5.5	4.1	3.2	-2.0	4.9	5.9	31.5	27.8	27.3	32.7	2.5	10	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	73.7	60.0	69.7	58.3	66.1	56.4	62.2	71.0	59.5	67.2	57.6	64.1	7.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	72.2	66.2	55.9	66.7	62.3	54.2	62.7	60.1	27.9	71.6	26.1	67.4	24.9	64.1	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.2	19.8	17.2	-6.1	80.2	2.7	5.6	-8.0	84.2	-9.6	87.5	-11.1	90.6	-13.0	94.6		
(5)			-6.2	65.1	2.8	3.2	-8.3	67.4	-9.9	69.3	-11.5	71.1	-13.6	73.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	23.4	24.2	27.6	35.3	44.4	50.4	56.3	54.2	47.9	38.9	31.5	27.5		
	DBStd	13.49	9.46	9.22	7.88	5.59	6.59	5.84	5.77	4.85	5.54	6.85	8.26	8.81		
	HDD50	4664	823	722	694	441	201	63	5	14	102	347	554	698		
	HDD65	9661	1288	1142	1159	891	640	438	280	335	513	808	1004	1163		
	CDD50	490	0	0	0	0	26	76	200	145	39	4	0	0		
	CDD65	12	0	0	0	0	0	1	10	1	0	0	0	0		
	CDH74	138	0	0	0	0	10	17	102	9	0	0	0	0		
	CDH80	23	0	0	0	0	1	2	19	1	0	0	0	0		
	WSAvg	7.4	7.7	8.0	7.6	7.5	8.2	7.9	7.3	6.7	6.7	6.8	6.8	7.5		
	PrecAvg	50.4	4.7	4.4	4.6	2.7	2.8	2.7	3.5	3.9	5.8	5.5	4.4	5.2		
	PrecMax	62.1	12.3	8.9	10.6	6.3	4.5	5.0	7.2	9.1	12.1	10.5	8.9	12.3		
(15) Precipitation	PrecMin	41.7	0.0	0.8	0.7	0.4	0.6	0.7	0.8	1.7	1.5	1.2	0.3	0.0		
	PrecStd	6.3	3.0	2.0	2.6	1.5	1.2	1.1	2.0	1.7	2.7	2.2	2.6	3.0		
	DB	44.4	42.8	45.8	57.5	73.0	75.6	82.3	74.1	67.0	58.8	48.6	45.5			
	MCWB	41.8	39.4	38.7	46.1	56.0	58.6	65.8	61.4	55.4	49.8	44.9	42.7			
	2% DB	41.3	41.0	42.2	51.4	65.9	69.8	76.3	69.5	62.5	53.8	46.5	43.1			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	38.6	37.9	37.7	42.5	52.1	56.4	62.5	59.2	54.1	47.3	43.3	39.9			
	DB	39.1	39.2	40.2	46.5	59.3	64.7	71.5	66.0	58.9	50.4	44.6	40.8			
	MCWB	36.5	36.8	36.3	40.3	48.8	53.7	60.1	57.5	52.2	46.8	41.2	37.4			
	10% DB	36.0	37.2	38.2	43.7	55.1	60.5	67.6	62.8	55.8	48.4	42.4	38.3			
	MCWB	33.4	34.7	35.1	38.6	46.1	51.9	58.4	56.3	51.1	45.3	39.1	35.6			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	42.1	41.1	40.3	46.8	56.5	59.9	67.3	64.1	57.1	50.9	46.7	44.0			
	MCDB	43.0	41.9	43.0	56.1	72.3	72.3	79.6	71.9	63.1	54.9	47.9	44.6			
	2% WB	39.1	38.4	38.7	43.6	52.2	57.4	64.1	60.3	54.9	49.2	44.4	40.3			
	MCDB	40.7	39.8	41.0	47.9	63.9	67.2	74.6	66.9	59.9	52.1	45.8	42.5			
	5% WB	36.5	36.9	36.9	41.3	49.5	55.0	61.2	58.6	53.5	47.5	41.5	37.7			
(31) Mean Daily Temperature Range	MCDB	38.5	38.8	39.3	45.2	58.8	62.6	69.5	64.7	57.8	50.3	43.5	40.0			
	10% WB	34.0	34.7	35.2	39.4	46.9	52.8	58.9	56.7	52.0	45.6	39.2	35.6			
	MCDB	35.6	36.6	37.8	43.1	53.2	59.2	65.7	61.6	55.6	48.1	41.5	37.9			
	5% DB	11.1	11.7	12.7	12.0	12.8	11.5	12.5	12.3	11.6	9.4	9.4	10.5			
	MCDBR	9.9	9.6	9.7	15.4	21.2	20.4	22.2	19.2	16.1	9.9	8.8	9.9			
(36) Clear-Sky Solar Irradiance	MCWBR	8.4	8.4	7.1	9.5	10.3	9.6	10.6	10.3	9.3	7.2	7.1	8.4			
	5% WB	9.6	9.4	9.3	13.4	19.8	17.1	20.6	17.0	13.5	9.0	8.7	9.5			
	MCDBR	8.8	8.9	7.4	9.1	10.0	9.0	10.7	9.7	8.5	7.4	7.8	8.6			
	taub	N/A	0.262	0.303	0.324	0.336	0.338	0.343	0.332	0.313	0.276	0.353	N/A			
	taud	N/A	2.106	2.181	2.285	2.428	2.495	2.510	2.528	2.551	2.278	2.655	N/A			
(42) All-Sky Solar Radiation	Ebn at Noon	N/A	162	227	260	272	274	268	259	234	159	32	0			
	Edh at Noon	N/A	14	25	30	30	29	27	24	18	12	3	0			
	RadAvg	7	101	450	1029	1525	1546	1400	1004	537	183	23	0			
(45)	RadStd	1	22	73	87	128	208	152	95	58	33	2	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page